

A10 ARPAX TOILET CONCENTRATE
Section: 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1 Product identifier: **A10 ARPAX TOILET CONCENTRATE**
Substance type: CLP Mixture

1.2 Relevant identified uses of the substance or mixture and uses advised against:

Use of the Substance/Mixture : CLEANER

Recommended restrictions on use : Reserved for industrial and professional use.

1.3 Details of the supplier of the safety data sheet:

COMPANY IDENTIFICATION

Ecolab Ltd.
PO Box 11; Winnington Avenue
Northwich, Cheshire., CW8 4DX, United Kingdom
TEL: + 44 (0)1606 74488

LOCAL COMPANY IDENTIFICATION

Ecolab Ltd.
PO Box 11; Winnington Avenue
Northwich, Cheshire., CW8 4DX, United Kingdom
TEL: + 44 (0)1606 74488

For Product Safety information please contact: msdseame@nalco.com

1.4 Emergency telephone number:

Emergency telephone number : Trans-European
+441618841235
+32-(0)3-575-5555 Trans-European Address European
Economic Area HQ

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Section: 2. HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Flammable liquids, Category 3	H226
Skin corrosion, Category 1	H314
Serious eye damage, Category 1	H318
Acute aquatic toxicity, Category 1	H400
Chronic aquatic toxicity, Category 2	H411

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

Hazard pictograms :



Signal Word : Danger

Hazard Statements :	H226	Flammable liquid and vapour.
	H314	Causes severe skin burns and eye damage.
	H400	Very toxic to aquatic life.
	H411	Toxic to aquatic life with long lasting effects.

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Precautionary Statements : **Prevention:**

P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

P273 Avoid release to the environment.

P280 Wear protective gloves/ eye protection/ face protection.

Response:

P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower.

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P310 Immediately call a POISON CENTER/doctor.

Hazardous components which must be listed on the label:

2,2'-(octadec-9-enylimino)bisethanol

2.3 Other hazards

Do not mix with bleach or other chlorinated products – will cause chlorine gas.

Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS**3.2 Mixtures****Hazardous components**

Chemical Name	CAS-No. EC-No. REACH No.	Classification (REGULATION (EC) No 1272/2008)	Concentration: [%]
Citric Acid, Monohydrate	5949-29-1 201-069-1 01-2119457026-42	Eye irritation Category 2; H319	20 - < 25
2,2'-(octadec-9-enylimino)bisethanol	25307-17-9 246-807-3 01-2119510876-35	Acute toxicity Category 4; H302 Skin corrosion Category 1B; H314 Serious eye damage Category 1; H318 Acute aquatic toxicity Category 1; H400 Chronic aquatic toxicity Category 1; H410	10 - < 20
Coconut oil derivatives	90170-43-7 290-476-8 01-2119976233-35	Eye irritation Category 2; H319	5 - < 10
Sulfamic Acid	5329-14-6 226-218-8 01-2119488633-28	Skin irritation Category 2; H315 Eye irritation Category 2; H319 Chronic aquatic toxicity Category 3; H412 According to OECD 404 and 405 the irritancy threshold of sulphamic acid for skin and eyes is above 10%	3 - < 5
propanaminium-1, amino-3 N-(carboxyméthyl) N,N-diméthyl-, dérivés N-acyles en C8-18, hydroxydes, sels internes	97862-59-4	Serious eye damage Category 1; H318	4 - < 5
Isopropanol	67-63-0 200-661-7 01-2119457558-25	Flammable liquids Category 2; H225 Eye irritation Category 2; H319 Specific target organ toxicity - single exposure Category 3; H336	3 - < 5

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Sodium Octyl Sulfate	142-31-4 205-535-5 01-2119966154-35	Serious eye damage/eye irritation Category 1; H318 Skin corrosion/irritation Category 2; H315 Skin irritation Category 2; H315 Serious eye damage Category 1; H318	1 - < 2.5
Linear(C12-C14)alkanol, ethoxylated, sulfated, sodium salt	68891-38-3 500-234-8 01-2119488639-16	Skin irritation Category 2; H315 Serious eye damage Category 1; H318 Chronic aquatic toxicity Category 3; H412	1 - < 2.5

For the full text of the H-Statements mentioned in this Section, see Section 16.

Section: 4. FIRST AID MEASURES**4.1 Description of first aid measures**

- If inhaled : Remove to fresh air.
Treat symptomatically.
Get medical attention if symptoms occur.
- In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes.
Use a mild soap if available.
Wash clothing before reuse.
Thoroughly clean shoes before reuse.
Get medical attention immediately.
- In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.
Remove contact lenses, if present and easy to do. Continue rinsing.
Get medical attention immediately.
- If swallowed : Rinse mouth with water.
Do NOT induce vomiting.
Never give anything by mouth to an unconscious person.
Get medical attention immediately.
- Protection of first-aiders : In event of emergency assess the danger before taking action.
Do not put yourself at risk of injury. If in doubt, contact emergency responders. Use personal protective equipment as required.

4.2 Most important symptoms and effects, both acute and delayed

See Section 11 for more detailed information on health effects and symptoms.

4.3 Indication of immediate medical attention and special treatment needed

- Treatment : Treat symptomatically.

Section: 5. FIREFIGHTING MEASURES**5.1 Extinguishing media**

- Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

5.2 Special hazards arising from the substance or mixture

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- | | |
|--------------------------------------|---|
| Specific hazards during firefighting | : Not flammable or combustible. |
| Hazardous combustion products | : Depending on combustion properties, decomposition products may include following materials:
Carbon oxides
nitrogen oxides (NOx)
Sulphur oxides |

5.3 Advice for firefighters

- | | |
|---|--|
| Special protective equipment for firefighters | : Use personal protective equipment. |
| Further information | : Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes. |

Section: 6. ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

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|------------------------------------|---|
| Advice for non-emergency personnel | : Ensure adequate ventilation.
Keep people away from and upwind of spill/leak.
Avoid inhalation, ingestion and contact with skin and eyes.
When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
Ensure clean-up is conducted by trained personnel only.
Refer to protective measures listed in sections 7 and 8. |
| Advice for emergency responders | : If specialised clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials. |

6.2 Environmental precautions

- | | |
|---------------------------|--|
| Environmental precautions | : Do not allow contact with soil, surface or ground water. |
|---------------------------|--|

6.3 Methods and materials for containment and cleaning up

- | | |
|-------------------------|--|
| Methods for cleaning up | : Stop leak if safe to do so.
Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).
Flush away traces with water.
For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway. |
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6.4 Reference to other sections

- See Section 1 for emergency contact information.
For personal protection see section 8.
See Section 13 for additional waste treatment information.

Section: 7. HANDLING AND STORAGE

A10 ARPAX TOILET CONCENTRATE**7.1 Precautions for safe handling**

- Advice on safe handling : Do not ingest. Do not breathe spray, vapour. Do not get in eyes, on skin, or on clothing. Wash hands thoroughly after handling. Use only with adequate ventilation. Do not mix with bleach or other chlorinated products – will cause chlorine gas.
- Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

7.2 Conditions for safe storage, including any incompatibilities

- Requirements for storage areas and containers : Keep in a cool, well-ventilated place. Keep away from strong bases. Keep out of reach of children. Keep container tightly closed. Store in suitable labelled containers.
- Suitable material : Keep in properly labelled containers.

7.3 Specific end uses

- Specific use(s) : CLEANER

Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION**8.1 Control parameters****Occupational Exposure Limits**

Components	CAS-No.	Value type (Form of exposure)	Control parameters	Basis
Isopropanol	67-63-0	OELV - 8 hrs (TWA)	200 ppm	IR_OEL
Further information	Sk	Substances which have the capacity to penetrate intact skin when they come in contact with it, and be absorbed into the body		
		OELV - 15 min (STEL)	400 ppm	IR_OEL
Further information	Sk	Substances which have the capacity to penetrate intact skin when they come in contact with it, and be absorbed into the body		

DNEL

Sulfamic Acid	:	End Use: Workers Exposure routes: Dermal Potential health effects: long term - systemic 10 mg/kg
Isopropanol	:	End Use: Workers Exposure routes: Dermal Potential health effects: Long-term systemic effects Value: 888 mg/cm ²
	:	End Use: Workers Exposure routes: Inhalation Potential health effects: Long-term systemic effects Value: 500 mg/m ³
	:	End Use: Consumers Exposure routes: Dermal Potential health effects: Long-term systemic effects

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		Value: 319 mg/cm ²
		End Use: Consumers Exposure routes: Inhalation Potential health effects: Long-term systemic effects Value: 89 mg/m ³
		End Use: Consumers Exposure routes: Ingestion Potential health effects: Long-term systemic effects Value: 26 ppm

PNEC

Sulfamic Acid	:	Fresh water Value: 0.048 mg/l
		Marine water Value: 0.0048 mg/l
		Intermittent release Value: 0.48 mg/l
		STP Value: 2 mg/l
		Fresh water sediment Value: 0.173 mg/kg
		Marine sediment Value: 0.0173 mg/kg
		Soil Value: 0.00638 mg/kg
Isopropanol	:	Fresh water Value: 140.9 mg/l
		Marine water Value: 140.9 mg/l
		Intermittent use/release Value: 140.9 mg/l
		Fresh water Value: 552 mg/kg
		Marine sediment Value: 552 mg/kg
		Soil Value: 28 mg/kg
		Sewage treatment plant Value: 2251 mg/l
		Oral Value: 160 mg/kg

8.2 Exposure controls**Appropriate engineering controls**

Effective exhaust ventilation system.

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Maintain air concentrations below occupational exposure standards.

Individual protection measures

- Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.
- Eye/face protection (EN 166) : Safety goggles
Face-shield
- Hand protection (EN 374) : Recommended preventive skin protection
Gloves
Nitrile rubber
butyl-rubber
Breakthrough time: 1 – 4 hours
Minimum thickness for butyl-rubber 0.7 mm for nitrile rubber 0.4 mm or equivalent (please refer to the gloves manufacturer/distributor for advice).
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.
- Skin and body protection (EN 14605) : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing including appropriate safety shoes
- Respiratory protection (EN 143, 14387) : When respiratory risks cannot be avoided or sufficiently limited by technical means of collective protection or by measures, methods or procedures of work organization, consider the use of certified respiratory protection equipment meeting EU requirements (89/656/EEC, (EU) 2016/425), or equivalent, with filter type: A-P

Environmental exposure controls

- General advice : Consider the provision of containment around storage vessels.

Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

- Appearance : viscous liquid
- Colour : clear, blue
- Odour : characteristic
- Flash point :
not determined
- pH : 1.5 - 2, (20 °C)
- Odour Threshold : no data available
- Melting point/freezing point : no data available
- Initial boiling point and boiling : no data available

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range

Evaporation rate : no data available

Flammability (solid, gas) : no data available

Upper explosion limit : no data available

Lower explosion limit : no data available

Vapour pressure : no data available

Relative vapour density : no data available

Relative density : 1.13 - 1.15

Solubility(ies)

Water solubility : soluble in cold water, soluble in hot water

Solubility in other solvents : no data available

Partition coefficient: n-octanol/water : no data available

Auto-ignition temperature : no data available

Thermal decomposition : no data available

Viscosity, dynamic : no data available

Viscosity, kinematic : no data available

Explosive properties : no data available

Oxidizing properties : no data available

9.2 Other information

no data available

Section: 10. STABILITY AND REACTIVITY

10.1 Reactivity

No dangerous reaction known under conditions of normal use.

10.2 Chemical stability

Stable under normal conditions.

10.3 Possibility of hazardous reactions

Hazardous reactions : Do not mix with bleach or other chlorinated products – will cause chlorine gas.

10.4 Conditions to avoid

Conditions to avoid : None known.

10.5 Incompatible materials

Materials to avoid : Strong bases

10.6 Hazardous decomposition products

Hazardous decomposition products : Depending on combustion properties, decomposition products may include following materials:
Carbon oxides
nitrogen oxides (NO_x)

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Sulphur oxides

Section: 11. TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Toxicity

Product

Acute oral toxicity : Acute toxicity estimate : > 2,000 mg/kg
Acute inhalation toxicity : There is no data available for this product.
Acute dermal toxicity : There is no data available for this product.
Skin corrosion/irritation : There is no data available for this product.
Serious eye damage/eye irritation : There is no data available for this product.
Respiratory or skin sensitization : There is no data available for this product.
Carcinogenicity : There is no data available for this product.
Reproductive effects : There is no data available for this product.
Germ cell mutagenicity : There is no data available for this product.
Teratogenicity : There is no data available for this product.
STOT - single exposure : There is no data available for this product.
STOT - repeated exposure : There is no data available for this product.
Aspiration toxicity : There is no data available for this product.

Components

Acute oral toxicity : Citric Acid, Monohydrate
LD50 rat: 11,700 mg/kg

2,2'-(octadec-9-enylimino)bisethanol
LD50 rat: 1,260 mg/kg

Sulfamic Acid
LD50 rat: 3,160 mg/kg

Isopropanol
LD50 rat: 5,840 mg/kg

Sodium Octyl Sulfate
LD50 rat: 2,175 mg/kg

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Linear(C12-C14)alkanol, ethoxylated, sulfated, sodium salt
LD50 rat: 3,350 mg/kg

Components

Acute inhalation toxicity : Isopropanol
LC50 rat: > 30 mg/l
Exposure time: 4 h
Test atmosphere: vapour

Components

Acute dermal toxicity : Citric Acid, Monohydrate
LD50 rat: > 2,000 mg/kg

Sulfamic Acid
LD50 rat: > 2,000 mg/kg

Isopropanol
LD50 rabbit: 12,870 mg/kg

Sodium Octyl Sulfate
LD50 rabbit: > 500 mg/kg

Linear(C12-C14)alkanol, ethoxylated, sulfated, sodium salt
LD50 rabbit: 8,000 mg/kg

Potential Health Effects

Eyes : Causes serious eye damage.

Skin : Causes severe skin burns.

Ingestion : Causes digestive tract burns.

Inhalation : May cause nose, throat, and lung irritation.

Chronic Exposure : Health injuries are not known or expected under normal use.

Experience with human exposure

Eye contact : Redness, Pain, Corrosion

Skin contact : Redness, Pain, Corrosion

Ingestion : Corrosion, Abdominal pain

Inhalation : Respiratory irritation, Cough

Further information : no data available

Section: 12. ECOLOGICAL INFORMATION

12.1 Ecotoxicity

Product

Environmental Effects : Very toxic to aquatic life. Toxic to aquatic life with long

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lasting effects.

Toxicity to fish : no data available

Toxicity to daphnia and other aquatic invertebrates : no data available

Toxicity to algae : no data available

Components

Toxicity to fish : Citric Acid, Monohydrate
96 h LC50 Fish: > 100 mg/l

2,2'-(octadec-9-enylimino)bisethanol
96 h LC50 Danio rerio (zebra fish): 0.1 mg/l

Coconut oil derivatives
96 h LC50 Oncorhynchus mykiss (rainbow trout): 4.2 mg/l

Isopropanol
96 h LC50 Pimephales promelas (fathead minnow):
9,640 mg/l

Linear(C12-C14)alkanol, ethoxylated, sulfated, sodium salt
96 h LC50 Fish: 7.1 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates : 2,2'-(octadec-9-enylimino)bisethanol
48 h EC50 Daphnia magna (Water flea): 0.043 mg/l

Coconut oil derivatives
48 h EC50 Daphnia magna (Water flea): 29 mg/l

Isopropanol
LC50 Daphnia magna (Water flea): > 10,000 mg/l

Sodium Octyl Sulfate
48 h EC50 Daphnia: 31 mg/l

Components

Toxicity to algae : 2,2'-(octadec-9-enylimino)bisethanol
72 h EC50 Pseudokirchneriella subcapitata (microalgae): 0.0538 mg/l

Coconut oil derivatives
72 h EC50 Chlorella vulgaris (Fresh water algae): 9.4 mg/l

Sulfamic Acid
72 h EC50: 48 mg/l

Components

Toxicity to bacteria : Isopropanol
1,050 mg/l

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Components

Toxicity to fish (Chronic toxicity) : Linear(C12-C14)alkanol, ethoxylated, sulfated, sodium salt
28 d NOEC Oncorhynchus mykiss (rainbow trout): 0.14 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : 2,2'-(octadec-9-enylimino)bisethanol
21 d EC50: 0.0463 mg/l

Coconut oil derivatives
21 d NOEC Daphnia magna (Water flea): 10 mg/l

12.2 Persistence and degradability

Product

no data available

Components

Biodegradability : Citric Acid, Monohydrate
Result: Readily biodegradable.

2,2'-(octadec-9-enylimino)bisethanol
Result: Readily biodegradable.

Coconut oil derivatives
Result: Readily biodegradable.

Sulfamic Acid
Result: Not applicable - inorganic

propanaminium-1, amino-3 N-(carboxyméthyl) N,N-diméthyl-,
dérivés N-acyles en C8-18, hydroxydes, sels internes
Result: no data available

Isopropanol
Result: Readily biodegradable.

Sodium Octyl Sulfate
Result: Biodegradable

Linear(C12-C14)alkanol, ethoxylated, sulfated, sodium salt
Result: Readily biodegradable.

12.3 Bioaccumulative potential

no data available

12.4 Mobility in soil

no data available

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12.5 Results of PBT and vPvB assessment

Product

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

12.6 Other adverse effects

no data available

Section: 13. DISPOSAL CONSIDERATIONS

Dispose of in accordance with the European Directives on waste and hazardous waste. Waste codes should be assigned by the user, preferably in discussion with the waste disposal authorities.

13.1 Waste treatment methods

Product : The product should not be allowed to enter drains, water courses or the soil.
Where possible recycling is preferred to disposal or incineration.
If recycling is not practicable, dispose of in compliance with local regulations.
Dispose of wastes in an approved waste disposal facility.

Contaminated packaging : Dispose of as unused product.
Empty containers should be taken to an approved waste handling site for recycling or disposal.
Do not re-use empty containers.

Guidance for Waste Code selection : Organic wastes containing dangerous substances. If this product is used in any further processes, the final user must redefine and assign the most appropriate European Waste Catalogue Code. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in compliance with applicable European (EU Directive 2008/98/EC) and local regulations.

Section: 14. TRANSPORT INFORMATION

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport (ADR/ADN/RID)

14.1 UN number:	UN 1987
14.2 UN proper shipping name:	ALCOHOL, N.O.S. (Isopropanol)
14.3 Transport hazard class(es):	3
14.4 Packing group:	III
14.5 Environmental hazards:	Yes
14.6 Special precautions for user:	Not applicable.

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14.1 UN number:	UN 1987
14.2 UN proper shipping name:	ALCOHOL, N.O.S. (Isopropanol)
14.3 Transport hazard class(es):	3
14.4 Packing group:	III
14.5 Environmental hazards:	Yes
14.6 Special precautions for user:	Not applicable.

Sea transport (IMDG/IMO)

14.1 UN number:	UN 1987
14.2 UN proper shipping name:	ALCOHOL, N.O.S. (Isopropanol)
14.3 Transport hazard class(es):	3
14.4 Packing group:	III
14.5 Environmental hazards:	Yes (Marine Pollutant)
14.6 Special precautions for user:	Not applicable.
14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code:	Not applicable.

Section: 15. REGULATORY INFORMATION**15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture:**

INTERNATIONAL CHEMICAL CONTROL LAWS

15.2 Chemical Safety Assessment:

No Chemical Safety Assessment has been carried out on the product.

Section: 16. OTHER INFORMATION**Procedure used to derive the classification according to REGULATION (EC) No 1272/2008**

Classification	Justification
Flammable liquids 3, H226	Based on product data or assessment
Skin corrosion 1, H314	Based on product data or assessment
Serious eye damage 1, H318	Based on product data or assessment
Acute aquatic toxicity 1, H400	Calculation method
Chronic aquatic toxicity 2, H411	Calculation method

Full text of H-Statements

H225	Highly flammable liquid and vapour.
H302	Harmful if swallowed.
H314	Causes severe skin burns and eye damage.
H315	Causes skin irritation.
H318	Causes serious eye damage.
H319	Causes serious eye irritation.
H336	May cause drowsiness or dizziness.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H412	Harmful to aquatic life with long lasting effects.

Full text of other abbreviations

ADN – European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR – European Agreement concerning the International Carriage of Dangerous Goods by

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Road; AICS – Australian Inventory of Chemical Substances; ASTM – American Society for the Testing of Materials; bw – Body weight; CLP – Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR – Carcinogen, Mutagen or Reproductive Toxicant; DIN – Standard of the German Institute for Standardisation; DSL – Domestic Substances List (Canada); ECHA – European Chemicals Agency; EC-Number – European Community number; EC_x – Concentration associated with x% response; EL_x – Loading rate associated with x% response; EmS – Emergency Schedule; ENCS – Existing and New Chemical Substances (Japan); ErC_x – Concentration associated with x% growth rate response; GHS – Globally Harmonized System; GLP – Good Laboratory Practice; IARC – International Agency for Research on Cancer; IATA – International Air Transport Association; IBC – International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC₅₀ – Half maximal inhibitory concentration; ICAO – International Civil Aviation Organization; IECSC – Inventory of Existing Chemical Substances in China; IMDG – International Maritime Dangerous Goods; IMO – International Maritime Organization; ISHL – Industrial Safety and Health Law (Japan); ISO – International Organisation for Standardization; KECI – Korea Existing Chemicals Inventory; LC₅₀ – Lethal Concentration to 50 % of a test population; LD₅₀ – Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL – International Convention for the Prevention of Pollution from Ships; n.o.s. – Not Otherwise Specified; NO(A)EC – No Observed (Adverse) Effect Concentration; NO(A)EL – No Observed (Adverse) Effect Level; NOELR – No Observable Effect Loading Rate; NZIoC – New Zealand Inventory of Chemicals; OECD – Organization for Economic Co-operation and Development; OPPTS – Office of Chemical Safety and Pollution Prevention; PBT – Persistent, Bioaccumulative and Toxic substance; PICCS – Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR – (Quantitative) Structure Activity Relationship; REACH – Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID – Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT – Self-Accelerating Decomposition Temperature; SDS – Safety Data Sheet; TCSI – Taiwan Chemical Substance Inventory; TRGS – Technical Rule for Hazardous Substances; TSCA – Toxic Substances Control Act (United States); UN – United Nations; vPvB – Very Persistent and Very Bioaccumulative

Sources of key data used to compile the Safety Data Sheet : IARC Monographs on the Evaluation of the Carcinogenic Risk of Chemicals to Man, Geneva: World Health Organization, International Agency for Research on Cancer.

The possible key literature references and data sources which may have been used in conjunction with the consideration of expert judgment to compile this Safety Data Sheet: European regulations/directives (including (EC) No. 1907/2006, (EC) No. 1272/2008), supplier data, inter-net, ESIS, IUCLID, ERICards, Non European official regulatory data and other data sources.

Prepared By : Regulatory Affairs

Numbers quoted in the MSDS are given in the format: 1,000,000 = 1 million and 1,000 = 1 thousand. 0.1 = 1 tenth and 0.001 = 1 thousandth

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.